

# Supply Chain Analytics: Phase 1 – Data Validation & Cleaning

## Overview

This phase focuses on establishing a **"Single Source of Truth"** by auditing the raw supply chain database. We identify structural inconsistencies, validate relational integrity, and develop a cleaning layer using SQL views to prepare the data for advanced risk modeling.

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## 1. Data Audit & Integrity Checks

Before transformation, we performed a rigorous audit of the raw schema to identify "data leakage" and orphaned records.

### Row Count Baseline

We established initial baselines across all core tables to ensure data ingestion was complete:

- **Entities:** Customers, Products, Suppliers, Warehouses.
- **Transactions:** Purchase Orders, PO Items, Shipments, Inventory Levels.

### Relational Integrity (Foreign Key Validation)

We conducted `left join` audits to identify orphaned records. These are critical "silent killers" in supply chain data that lead to incorrect inventory totals.

- **Inventory Audit:** Verified that every inventory record maps to a valid warehouse and product.
- **Procurement Audit:** Verified that all POs and PO Items are linked to active suppliers and products.
- **Logistics Audit:** Checked for shipments tied to non-existent POs or warehouses.

## 2. Transformation & Cleaning Layer (Views)

Rather than altering raw data, we implemented a **View-based Cleaning Layer**. This ensures that the original data remains intact while providing sanitized data for analysis.

## **purchase\_order\_cn**

- **Standardization:** Normalized order statuses to lowercase/trimmed strings.
- **Date Sanity:** Flagged records where expected\_delivery preceded the order\_date.
- **Financial Scrubbing:** Rounded amounts and handled NULL/Zero totals.

## **vw\_shipments\_clean**

- **Carrier Normalization:** Standardized carrier names to prevent duplication in reports (e.g., "UPS" vs "ups").
- **Cancellation Logic:** Automatically flags shipments tied to canceled POs to prevent "ghost inventory" projections.

## **inventory\_cn**

- **Dynamic Risk Flagging:** Created the is\_below\_reorder\_level logic by comparing live quantity\_on\_hand against the product reorder\_level.
- **Constraint Enforcement:** Excluded negative inventory counts that represent data entry errors.

## **vw\_products\_clean & vw\_suppliers\_clean**

- **Deduplication:** Utilized DISTINCT logic to ensure product-level analysis isn't skewed.
- **Standardization:** Normalized order statuses to trimmed strings (e.g product name,description,SKU)
- **Status Mapping:** Standardized "Suspended" or "Active" labels for suppliers to enable risk filtering.
- **Flagging numerical anomalies :** (e.g p.reorder\_level < 0, p.unit\_price IS NULL OR p.unit\_price <= 0)

# **3. Preliminary Findings**

## **Data Quality & Integrity Issues Identified**

- **Orphaned Records Detected**
  - 2 purchase order records were excluded due to missing or invalid references.

- **14** shipment records were excluded because they were not linked to a valid purchase order ID or warehouse ID.
- **24** invalid records were identified in the purchase order items table.
- **64** records were excluded from the inventory table due to data integrity issues.
- **Reference Data Validation**
  - No records were removed from the **suppliers** or **products** tables.
  - However, **suspended suppliers** and **suspicious product records** were flagged for monitoring rather than excluded, to preserve analytical visibility.
- **Date Consistency Issues**
  - **38** records with invalid or inconsistent date values were identified and flagged for downstream handling.
- **Supplier Risk Indicators**
  - **7** suppliers with a suspended status were flagged to track ongoing transactional activity and potential operational risk.
- **Inventory Risk Signals**
  - **5** critical stockout cases were detected, indicating immediate replenishment or supplier attention is required.

## 4. Next Steps

With the data sanitized and the relational integrity verified, the project will move to **Phase 2: Performance Metrics**, where we will calculate:

1. Lead Time Variability.
2. Supplier On-Time Delivery (OTD) rates.
3. Carrier reliability scores.